

Problem 1 (11.4.1)

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle$$

In coordinate representation

$$\psi(x,t) = e^{-i\hat{H}t/\hbar} \psi(x,0)$$

Suppose

$$\langle x | \pi | \psi \rangle = \pi \psi(x,0) = \psi(-x)$$

Then

$$\pi \psi(x,t) = \pi e^{-i\hat{H}t/\hbar} \psi(x,0)$$

However since $[\pi, \hat{H}] = 0$

$$\pi e^{-i\hat{H}t/\hbar} = e^{-i\hat{H}t/\hbar} \pi$$

Thus

$$\begin{aligned} \pi \psi(x,t) &= e^{-i\hat{H}t/\hbar} \pi \psi(x,0) \\ &= e^{-i\hat{H}t/\hbar} \psi(-x,0) \\ &= \psi(-x,t) \end{aligned}$$

The proof is identical if $\pi \psi(x,0) = -\psi(-x)$

Problem 2 (12.2.1)

$$U[R]|x, y\rangle = |x - y\varepsilon_z, x\varepsilon_z + y\rangle$$

This means

$$\langle x, y | U^\dagger[R] = \langle x - y\varepsilon_z, x\varepsilon_z + y |$$

However

$$U^\dagger(R) = 1 + \frac{i\varepsilon_z L_z}{\hbar}$$

Since $L_z = L_z^\dagger$

Thus

$$\langle x, y | U[R] = \langle x + y\varepsilon_z, x\varepsilon_z + y |$$

Thus

$$\langle x, y | U[R] |\psi\rangle = \psi(x + y\varepsilon_z, y - x\varepsilon_z)$$

Problem 3 (12.2.3)

$$\begin{aligned}x &= r \cos \phi \\y &= r \sin \phi\end{aligned}$$

$$\begin{aligned}\tan \phi &= \frac{y}{x} \\r^2 &= x^2 + y^2. \quad \frac{\partial r}{\partial x} = \frac{x}{r}\end{aligned}$$

Thus

$$\sec^2 \phi \, d\phi = \frac{dy}{x} - \frac{y dx}{x^2}$$

$$\frac{\partial \phi}{\partial y} = \frac{1}{x} \frac{x^2}{r^2} = \frac{x}{r^2}$$

$$\frac{\partial \phi}{\partial x} = -\frac{y}{x^2} \frac{x^2}{r^2} = -\frac{y}{r^2}$$

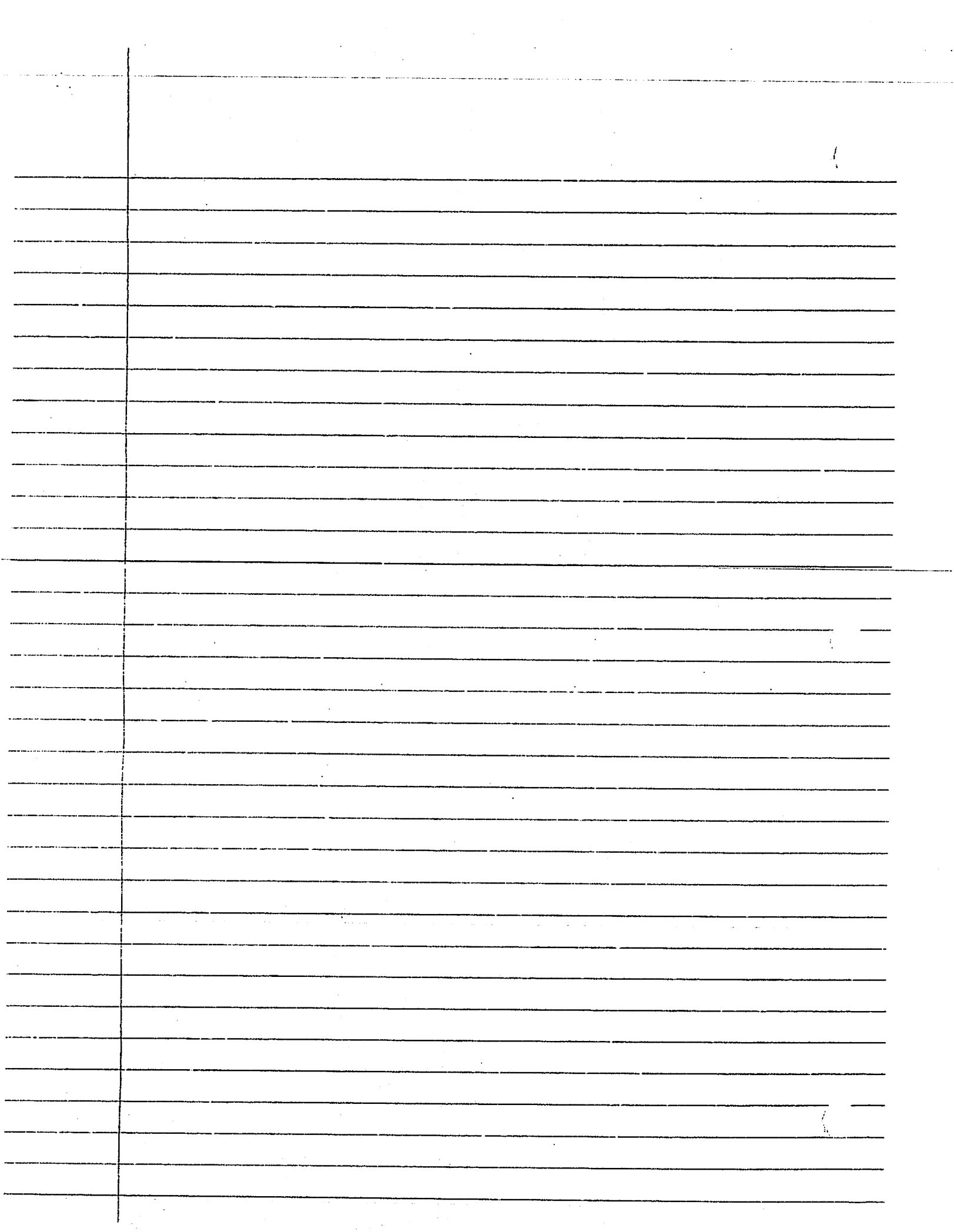
Thus

$$\begin{aligned}x \partial_y - y \partial_x &= x \left[\frac{x}{r^2} \partial_\phi + \frac{y}{r} \partial_r \right] \\&\quad - y \left[-\frac{y}{r^2} \partial_\phi + \frac{x}{r} \partial_r \right] \\&= \frac{x^2 + y^2}{r^2} \partial_\phi = \partial_\phi\end{aligned}$$

$$\Rightarrow L_z = -i\hbar \partial_\phi$$

By direct method this follows since
to lowest order

$$\begin{aligned}\psi(\rho, \phi - \epsilon_z) &= \left(1 - \epsilon_z \frac{\partial}{\partial \phi} \right) \psi(\rho, \phi) \\&= \left(1 - \frac{i}{\hbar} (-i\hbar \frac{\partial}{\partial \phi}) \right) \psi(\rho, \phi)\end{aligned}$$



Problem 4 (12.2.4)

$$\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{\varepsilon_x} \begin{pmatrix} x + \varepsilon_x \\ y \end{pmatrix} \xrightarrow{\varepsilon_z}$$

$$\begin{bmatrix} (x + \varepsilon_x) \cos \varepsilon_z - y \sin \varepsilon_z \\ (x + \varepsilon_x) \sin \varepsilon_z + y \cos \varepsilon_z \end{bmatrix}$$

$$\longrightarrow -\varepsilon_x$$

$$\begin{pmatrix} x \cos \varepsilon_z + \varepsilon_x (\cos \varepsilon_z - 1) - y \sin \varepsilon_z \\ (x + \varepsilon_x) \sin \varepsilon_z + y \cos \varepsilon_z \end{pmatrix}$$

$$\longrightarrow -\varepsilon_z$$

$$\begin{bmatrix} (x \cos \varepsilon_z + \varepsilon_x (\cos \varepsilon_z - 1) - y \sin \varepsilon_z) \cos \varepsilon_z \\ \quad + ((x + \varepsilon_x) \sin \varepsilon_z + y \cos \varepsilon_z) \sin \varepsilon_z \\ - (x \cos \varepsilon_z + \varepsilon_x (\cos \varepsilon_z - 1) - y \sin \varepsilon_z) \sin \varepsilon_z \\ \quad + ((x + \varepsilon_x) \sin \varepsilon_z + y \cos \varepsilon_z) \cos \varepsilon_z \end{bmatrix}$$

Keep terms up to $O(\varepsilon_z^2)$

$$\begin{pmatrix} x + \frac{1}{2} \varepsilon_x \varepsilon_z^2 + O(\varepsilon_x \varepsilon_z^4) \\ y + \varepsilon_x \varepsilon_z + O(\varepsilon_x \varepsilon_z^3) \end{pmatrix}$$

Thus the RHS of (12.2.24) is modified to

$$U[R(-\epsilon_z \hat{k})] T(-\epsilon_x) U[R(\epsilon_z \hat{k})] T(\epsilon_x) \\ = T\left(\frac{1}{2} \epsilon_x \epsilon_z^2 \hat{i} + \epsilon_x \epsilon_z \hat{j}\right)$$

Expanding LHS

$$\left(1 + \frac{i}{\hbar} \epsilon_z L_z - \frac{1}{2\hbar^2} \epsilon_z^2 L_z^2\right) \left(1 + \frac{i}{\hbar} \epsilon_x P_x\right) \\ \left(1 - \frac{i}{\hbar} \epsilon_z L_z - \frac{1}{2\hbar^2} \epsilon_z^2 L_z^2\right) \\ \left(1 - \frac{i}{\hbar} \epsilon_x P_x\right) \\ = \left(1 + \frac{i}{\hbar} \epsilon_z L_z - \frac{1}{2\hbar^2} \epsilon_z^2 L_z^2 + \frac{i}{\hbar} \epsilon_x P_x \right. \\ \left. - \frac{1}{\hbar^2} \epsilon_x \epsilon_z L_z P_x - \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x\right) \\ \left(1 - \frac{i}{\hbar} \epsilon_z L_z - \frac{1}{2\hbar^2} \epsilon_z^2 L_z^2 - \frac{i}{\hbar} \epsilon_x P_x \right. \\ \left. - \frac{1}{\hbar^2} \epsilon_x \epsilon_z L_z P_x + \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x\right)$$

We keep terms which are $O(1)$
 $O(\epsilon_x)$ $O(\epsilon_z)$ $O(\epsilon_x \epsilon_z)$ $O(\epsilon_x \epsilon_z^2)$

$$O(\epsilon_x): \quad \frac{i}{\hbar} \epsilon_x P_x - \frac{i}{\hbar} \epsilon_x P_x = 0$$

$$O(\epsilon_z): \quad \frac{i}{\hbar} \epsilon_z L_z - \frac{i}{\hbar} \epsilon_z L_z = 0$$

$$\begin{aligned} O(\epsilon_x \epsilon_z): \quad & \frac{1}{\hbar^2} \epsilon_z \epsilon_x L_z P_x + \frac{1}{\hbar^2} \epsilon_z \epsilon_x P_x L_z \\ & - \frac{1}{\hbar^2} \epsilon_x \epsilon_z L_z P_x - \frac{1}{\hbar^2} \epsilon_x \epsilon_z L_z P_x \\ & = \frac{\epsilon_x \epsilon_z}{\hbar^2} [P_x, L_z] \end{aligned}$$

$$\begin{aligned} O(\epsilon_x \epsilon_z^2) &= \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x \\ &\quad - \frac{i}{\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x \\ &\quad + \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x \\ &\quad - \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 P_x L_z^2 \checkmark \\ &\quad + \frac{i}{\hbar^3} \epsilon_x \epsilon_z^2 L_z P_x L_z \\ &\quad - \frac{i}{2\hbar^3} \epsilon_x \epsilon_z^2 L_z^2 P_x \checkmark \end{aligned}$$

$$\begin{aligned} &= \epsilon_x \epsilon_z^2 \left[-\frac{i}{2\hbar^3} L_z^2 P_x - \frac{i}{2\hbar^3} P_x L_z^2 \right. \\ &\quad \left. + L_z P_x L_z \right] \end{aligned}$$

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Comparing with RHS must have

$$-2L_z P_x L_z + P_x L_z^2 + L_z^2 P_x = \hbar^2 P_x$$

Since the RHS is

$$1 - \frac{i}{\hbar} \left(\frac{\epsilon_x \epsilon_z^2}{2} P_x + \epsilon_x \epsilon_z P_y \right)$$

This means we must have

$$[L_z, [L_z, P_x]] = \hbar^2 P_x$$

$$\text{Since } [L_z, P_x] = i\hbar P_y$$

$$[L_z, P_y] = -i\hbar P_x$$

this is satisfied

Problem 5 (12.3.2)

$$\psi(\rho, \phi + 2\pi) = e^{\frac{2\pi i l_z}{\hbar}} \left[A e^{i\phi l_z / \hbar} + B e^{\frac{i\phi l_z'}{\hbar}} e^{\frac{2\pi i (l_z' - l_z)}{\hbar}} \right]$$

If $\psi(\rho, \phi + 2\pi)$ represents the same state as $\psi(\rho, \phi)$

$$\psi(\rho, \phi + 2\pi) = e^{i\alpha} \psi(\rho, \phi)$$

thus
$$e^{\frac{2\pi i}{\hbar} (l_z' - l_z)} = 1$$

$$\Rightarrow l_z' - l_z = m\hbar$$

$$m = 0, \pm 1, \pm 2, \pm 3, \dots$$

If $(l_z, -l_z)$ are both allowed values

$$2l_z = m\hbar \Rightarrow l_z = \frac{m}{2}\hbar$$

Problem 6 (12.3.3)

$$\begin{aligned}\cos^2 \phi &= \frac{1}{4} (e^{i\phi} + e^{-i\phi})^2 \\ &= \frac{1}{4} (e^{2i\phi} + e^{-2i\phi} + 2)\end{aligned}$$

Thus

$$\begin{aligned}P(l_z=0) : P(l_z=2\hbar) : P(l_z=-2\hbar) \\ = 4 : 1 : 1\end{aligned}$$

Remember
We get

$$\sum P = 1$$

$$P(l_z=0) = \frac{4}{6} = \frac{2}{3}$$

$$P(l_z=2\hbar) = P(l_z=-2\hbar) = \frac{1}{6}$$

Clearly the ϕ dependence is irrelevant